

Research paper

Enhancing construction project management competencies with AI-driven assistants: A dual perspective from academia and industry

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ABSTRACT

The adoption of assistants based on artificial intelligence (AI) in the construction industry is creating new opportunities and challenges for project management. Despite the potential of these tools, few studies examine how they can enhance project management skills. Therefore, this study examines construction managers' perceptions of the influence of AI assistants on their project management competencies. A systematic literature review of 206 documents identified 35 core competencies required for effective construction project management. These competencies were categorized based on their relevance to the planning, execution, monitoring, and closure phases of construction projects. Using this framework, a structured questionnaire was developed and administered to 135 managers from academia and industry who use AI tools in construction management. They rated on a five-point scale how much they believe AI tools can improve each competency. An exploratory factor analysis (EFA) was conducted to identify knowledge areas and AI-influenced competency domains that can be strengthened through the application of AI tools. The findings suggest that AI tools have a positive impact on technological innovation capacity, scheduling efficiency, resource allocation, and project reporting. While significant differences were observed in the perceived applicability of AI based on participants' roles and levels of experience, there was overall consensus on AI's important role in enhancing competencies for managing construction projects. This study contributes to the ongoing discussion about integrating AI into professional training and competency development in construction project management.

1. Introduction

Project management competencies are essential for the successful development of construction projects [1,2], ensuring improved efficiency, proactive risk mitigation, and the achievement of project objectives. However, despite the collaborative nature of the construction industry, ongoing issues persist, including poor coordination [3], ineffective communication [4,5], and suboptimal decision-making processes [6]. These challenges often lead to delays and cost overruns [7–10]. The multidisciplinary nature of construction teams (comprising engineers, contractors, architects, and project managers) exacerbates these problems, as timely and accurate information exchange is crucial throughout the project lifecycle [11–13]. In Chile, for example, the National Council for Project Evaluation (CNEP) estimates that weaknesses in project

management during pre-construction phases can account for 4 % to 22 % of total project costs. This highlights the critical need for managers to possess a strong set of competencies in order to perform effectively in uncertain and complex environments. Shortcomings in planning and design during the early stages can lead to significant inefficiencies and budget issues during execution [14,15]. While structured meetings, digital documentation, and standardized management frameworks have improved multidisciplinary collaboration and project management, these conventional tools can be inadequate for the challenges posed by modern projects [16–18]. As a result, emerging technologies, particularly tools based on artificial intelligence (AI), are being explored as practical solutions to help construction managers enhance collaboration, accelerate decision-making, and improve information flow among project stakeholders [19–21]. Within this technological landscape,

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conversational AI assistants like ChatGPT are noteworthy for their ability to provide accurate answers, technical recommendations, and context-aware support in dynamic on-site environments [22].

The increasing integration of AI across various industrial sectors is transforming the nature of work by automating routine tasks, streamlining information flows, and enabling data-driven decision-making based on predictive analytics [23]. In the realm of construction project management, cutting-edge AI applications are already being used for tasks such as schedule generation, clash detection, and rapid scenario analysis [24]. These tools provide managers with real-time data that was previously inaccessible. General-purpose generative AI agents, such as Copilot, ChatGPT, Gemini, and DeepSeek, can be tailored to serve as domain-specific advisors. They analyze large document repositories, synthesize contractual obligations, and recommend alternative actions in seconds. By utilizing natural language processing, these agents mine communication logs, detect emerging sentiment patterns, and identify ambiguities, which helps prevent costly misunderstandings within multidisciplinary teams [25]. Additionally, machine learning models trained on historical project data can identify hidden risk factors and suggest preventive measures, fostering a culture of proactive problem-solving [26]. Importantly, these AI tools integrate diverse information generated throughout the design-build-operate continuum, improving transparency, traceability, and coordination across different disciplines [27]. However, there is limited understanding of which managerial competencies are most enhanced by AI, and how perceptions of this potential vary between current managers and the next generation of construction managers. Hence, it is crucial to address this knowledge gap if both academia and industry aim to create training pathways that leverage AI to develop the skills required for Construction 4.0.

Despite the undeniable promise of algorithm-driven tools, the adoption of AI in construction project management is hindered by some persistent barriers: organizational resistance to change, a lack of industry-wide implementation standards, and a shortage of professionals with the specialized training necessary to use these technologies effectively [28]. Nevertheless, task-oriented conversational assistants provide a low-barrier, highly adaptable way to integrate digital capabilities into traditionally analog environments. This enables companies to experiment incrementally, thereby reducing the friction often associated with sector-wide digital transformation. When implemented on a larger scale, AI has the potential to deliver significant benefits, including more timely, evidence-based decisions, improved coordination and collaboration, reduced accident rates through better risk prediction, and personalized upskilling for project stakeholders [29]. Therefore, it is crucial to understand how construction managers perceive these advantages and identify which competencies they believe can be enhanced through the use of AI. It is also essential to differentiate between the perspectives of incumbent managers who have adopted AI independently and those of student managers who have been educated in programs that already incorporate this technology. This distinction will help us understand how education and practice can align to develop the data-centric, adaptive, and collaborative mindset required for Construction 4.0 [30,31].

Existing research on the impact of artificial intelligence (AI) in project management has examined a wide range of themes. These include AI-enabled models that optimize scheduling and resource allocation [32–34], studies on process automation and its effects on operational efficiency in construction [35–37], assessments of AI-supported strategic decision-making [38–40], and investigations into AI's role in enhancing communication and coordination among multidisciplinary teams [41,42]. However, there is limited understanding of how these technologies are reshaping the competency profile required for construction project managers. Specifically, the literature lacks substantial evidence regarding (1) which managerial competencies can realistically be enhanced by AI-driven tools, (2) how the perceptions of practitioners and future professionals align or differ in this regard, and (3) what the competency domains are for managing construction projects influenced

by AI-based tools. To address this gap, the present study compares the perspectives of practicing construction managers who have voluntarily adopted AI applications with those of construction-related students who have been exposed to such tools in recently updated curricula. Through a survey designed for both groups, this study quantifies how AI adoption is perceived to strengthen key competencies (such as strategic decision-making, risk analysis, and real-time operational optimization) and how it may facilitate a shift from traditional management methods to data-driven, predictive, and proactive practices. The statistical analysis of the responses reveals significant trends and provides actionable guidelines for integrating emerging AI technologies into competency-based frameworks for construction project management.

2. Literature background

This chapter provides the literature background for the key constructs of the study. First, it synthesizes primary studies and recent advancements in construction management competencies, focusing on the essential skills and attributes required of construction project managers (see Section 2.1). Second, it reviews the most relevant applications of Artificial Intelligence (AI) in supporting construction project management. This section highlights how AI tools are used to enhance efficiency, decision-making, and overall project performance (see Section 2.2).

2.1. Evidence from the literature of managerial skills in construction

Recent studies have focused on identifying the competencies required by construction managers to lead projects effectively. These managers need a diverse set of skills to navigate the complexities of modern projects. Key competencies include leadership and management, project management, technical skills, safety, and compliance. Leadership and people management are critical competencies for construction managers. Effective leadership entails guiding project teams, making informed strategic decisions, and cultivating a collaborative work environment. According to Ghorbani [43], successful construction project managers exhibit strong leadership profiles, including the ability to motivate and manage teams, resolve conflicts, and maintain high morale. Pariafsai and Behzadan [44] emphasize the importance of interpersonal skills, such as communication and negotiation, which are essential for managing diverse teams and stakeholders. Additionally, Mireku et al. [45] emphasize the importance of emotional intelligence and adaptability, enabling managers to navigate complex project dynamics and maintain productive relationships. Tezel and Irlayici [46] identify communication, problem-solving, and decision-making as critical skills for construction managers. These skills are essential for guiding project teams and making strategic decisions. Yap and Chow [47] identify the project manager's and team's competencies as critical to project success. Effective leadership involves guiding teams and ensuring stakeholder engagement. Furthermore, effective communication and relationship management are crucial for the success of construction projects. Kissi et al. [48] underscore the importance of clear and consistent communication with stakeholders, including clients, contractors, and team members. Pariafsai et al. [49] identify communication skills as a critical competency, enabling managers to convey project goals, expectations, and progress effectively. Kissi et al. [50] emphasize the importance of interpersonal skills in establishing trust and promoting collaboration among project participants. Strong relationship management skills help construction managers navigate conflicts, negotiate agreements, and maintain positive working relationships. Oke et al. [51] found that communication, organizational, and budgeting skills are the top skills exhibited by project managers.

Project management and technical skills are foundational for construction managers. Technical knowledge encompasses understanding construction methods, materials, and technologies like Building Information Modeling (BIM) [52,53], which enhances project planning and

coordination. Safety management involves comprehensive risk assessments and fostering a strong safety culture to prevent accidents. Effective budget control ensures projects stay within financial constraints, requiring meticulous planning and monitoring of expenses. Ngo and Hwang [54] emphasize the importance of project communication, cost, and risk management in managing projects that utilize innovative technologies. Effective project management involves planning, scheduling, budgeting, and risk management. Ghorbani [43] identifies critical project management competencies, such as time management, cost control, and quality assurance. Technical skills, including knowledge of construction methods, materials, and technologies, are equally important. Pariafsai et al. [55] note that construction managers must be proficient in using project management software and understanding technical specifications to ensure project success. Arvanitis et al. [56] further stress the importance of digital literacy and the ability to leverage emerging technologies, such as Building Information Modeling (BIM) and data analytics, to enhance project outcomes. Ogunsanmi [57] highlights the importance of site management and planning, project management/administration, and health and safety management. Dobrucali et al. [58] emphasize that time management and cost management are foundational for effective project management. Construction managers must be proficient in these areas to ensure project success. Zhong and Chen [59] emphasize the importance of effective planning and control, as well as realistic cost and time estimates, in managing construction projects. Strategic thinking and analytical skills are crucial for construction managers to make informed decisions and drive project success. Rahaman et al. [60] emphasize the importance of strategic planning and the ability to analyze project data to identify potential risks and opportunities. Pariafsai et al. [61] discuss the need for construction managers to develop and implement effective project strategies, considering factors such as resource allocation, scheduling, and cost management. Asante et al. [62] highlight the role of data analysis and decision-making tools in optimizing project performance and achieving strategic objectives. Moshood et al. [63] emphasize that strategic thinking and analytical skills are crucial for making informed decisions and driving project success.

Safety and compliance are paramount in construction management. Ensuring the safety of workers and compliance with regulations is a key responsibility of construction managers. Sadeghi et al. [64] emphasize the importance of safety management in preventing accidents and fostering a culture of safety on construction sites. Pariafsai et al. [65] discuss the importance of understanding health and safety regulations and implementing effective safety protocols. Kissi et al. [66] emphasize the need for continuous training and development in safety practices to keep up with evolving industry standards and ensure a safe working environment. Yap et al. [67] highlight the importance of safety and quality management in construction projects. Ensuring compliance with safety regulations is essential for preventing accidents and maintaining a safe working environment. Furthermore, personal development and ethics are fundamental to the professional growth and integrity of construction managers. Sadeghi et al. [68] emphasize the importance of continuous learning and self-improvement, urging managers to stay current with industry trends and best practices. Pariafsai et al. [65] emphasize the role of ethical behavior in maintaining trust and credibility with stakeholders. Kissi et al. [69] emphasize the importance of construction managers upholding ethical standards, including honesty, transparency, and accountability, in all aspects of their work. Personal development initiatives, such as training programs and professional certifications, help construction managers enhance their skills and advance their careers. Akomea-Frimpong et al. [70] identify the ability to learn as a critical skill for construction managers. Demirkesen et al. [71] emphasize the importance of leadership and ethical practices, which are fundamental to maintaining trust and credibility with stakeholders.

2.2. Artificial intelligence applications in construction management

The growing complexity of construction projects, alongside the need for improved efficiency in processes and resource use, has led to increased interest in integrating AI with traditional project management tools [72,73]. AI technologies have shown significant potential to enhance various aspects of construction management, such as optimizing schedules, estimating costs, analyzing risks, controlling quality, and making decisions under uncertain conditions [74,75]. However, implementing AI in the construction sector is still considered an emerging topic. This is evident from the limited amount of academic research and the fragmented nature of current adoption practices. Existing studies highlight the benefits and challenges of integrating AI, including the necessity for high-quality data, workforce training, and alignment with established workflows. The following review presents some contributions from the literature, illustrating the current state of research and highlighting emerging opportunities for AI to assist construction project managers.

Aladag et al. [76] conducted a scoping review of 41 studies published between 2012 and 2023, using databases such as ASCE, IEEE Xplore, Web of Science, and Scopus. The primary aim of their review was to identify research focused on the application of AI in construction project management. The review highlighted the critical role of AI in enhancing time and cost management, emphasizing its potential to optimize key processes, including resource allocation, delay prediction, and quality control implementation. However, the authors noted that limited data availability across these sources poses a barrier to developing and validating robust AI models in this field. In a related study, Altaie and Dishar [77] surveyed 85 construction project engineers from both public and private sectors in Iraq to assess the integration of AI-based applications in Knowledge Management (KM) processes. The authors used statistical methods, including correlation and multiple regression analyses, to explore the relationship between the engineers' profiles and the perceived benefits of AI tools. The findings revealed a negative correlation with age and positive correlations with job positions and years of experience, suggesting that younger professionals and those in higher-ranking roles perceive greater advantages from AI integration. The study concluded that AI-driven KM processes can enhance decision-making, improve project planning, and strengthen risk management capabilities. Complementing these findings, Bushuyev and Ivko [78] proposed an integrative project management model that combines traditional methodologies with AI-based systems. They developed an algorithm that outlines the conceptual framework of this approach, which includes elements such as collaborative network analysis, resource optimization, and the establishment of performance indicators. The model demonstrated enhanced efficiency in resource allocation and decision-making, resulting in reduced response times in critical situations and improved decision quality across various project stages.

Khan et al. [79] examined the role of knowledge management in facilitating the adoption of AI within small and medium-sized enterprises (SMEs) in the construction sector. Drawing on the Diffusion of Innovation Theory and Organizational Knowledge Theory, the authors proposed a conceptual model that explores how openness to innovation and knowledge integration influence the readiness of SMEs to adopt AI tools. The results indicated that SMEs with strong knowledge integration practices and a culture that supports innovation were more likely to implement AI-based solutions. However, some barriers were also identified, including limited digital infrastructure and resistance to change. The authors concluded that fostering an innovation-oriented organizational culture and enhancing knowledge management capabilities are critical enablers for successful AI adoption in the construction industry. Lai et al. [80] conducted a scoping review to assess the current state of AI applications in construction management. The review analyzed 60 studies focusing on AI's role in managing costs, time, quality, and safety. The aim was to identify trends, challenges, and opportunities within the field. The findings indicated that AI is primarily utilized during the

design and construction phases to optimize planning, control costs, and enhance on-site safety. The study also highlighted ongoing challenges, including the limited integration of AI into traditional management practices and the absence of clear implementation frameworks. Liu et al. [81] conducted a systematic and bibliometric review of 60 peer-reviewed articles to map the current state of AI applications in production, operations, and logistics management within the context of modular construction. Using keyword analysis and knowledge mapping techniques, the study identified key areas where AI has been applied, including schedule optimization, cost control, quality assurance, and safety management. The review shows that AI applications are most prevalent during the early design and construction phases, where techniques such as genetic algorithms, deep learning neural networks, and reinforcement learning have been employed to enhance efficiency and support data-driven decision-making.

Rampini and Cecconi [82] reviewed the adoption of AI in asset management within the construction industry. Their study analyzed 578 articles sourced from major academic databases. The findings revealed some areas of importance. In the field of energy management, AI techniques were utilized for forecasting energy consumption, controlling intelligent systems, and evaluating post-occupancy comfort. In the realm of condition assessment and operations, the application of deep learning algorithms, computer vision-based crack inspections, and predictive maintenance has improved accuracy while reducing time and resource consumption. Similarly, Quoc-Bao Vo et al. [83] conducted a systematic review of 94 journal articles published between 2000 and 2024 to evaluate the adoption of AI in construction management. Their focus was on how AI contributes to optimizing the "Iron Triangle" of project success, which includes quality, time, and cost. The results indicated a growing international trend toward AI integration, with applications in schedule optimization and quality control. The authors stressed that incorporating technologies such as machine learning and natural language processing could greatly enhance project accuracy and efficiency, ultimately reducing delays and improving cost control. In a different yet relevant context, Wang [84] developed an advanced AI-powered financial management system designed for the construction sector. A three-layer neural network was constructed for early financial risk warning, supported by a hybrid forecasting model optimized using an ant colony algorithm to fine-tune the weights of individual predictors. The results indicated that the early warning model increased prediction accuracy by approximately 15 % compared to traditional methods. Furthermore, the hybrid model, optimized through the ant colony algorithm, reduced forecasting errors by 20 %, enhancing the reliability of financial risk predictions in construction-related enterprises. Zhang [85] proposed an AI-based safety management system for construction projects. The developed platform enables real-time detection and early warning of safety violations using object detection algorithms and spatial interaction analysis to identify hazardous behaviors and non-compliant conditions on-site. Immediate alerts are generated and sent to safety personnel, allowing for prompt preventive actions.

Despite an increasing body of literature highlighting the potential of AI to transform various aspects of construction management (ranging from cost estimation and scheduling to safety monitoring and asset optimization), most studies to date have primarily focused on technical implementations, system architectures, and theoretical frameworks. While these contributions are important, a research gap exists in empirical studies that explore how construction professionals, both in industry and academia, perceive the usefulness of AI tools in enhancing their managerial competencies. Specifically, there is a lack of exploratory research examining the subjective evaluations, expectations, and readiness to adopt AI technologies among construction project managers and students in construction-related academic programs. Therefore, this study aims to fill this gap by investigating the perceived value of AI tools in improving project management skills and comparing the opinions of industry professionals with those of future practitioners.

3. Research method

The research method used to examine the perceived impact of Artificial Intelligence (AI) on the development of project management competencies in construction was organized into three sequential stages: (1) Identification of key competencies required for effective construction project management, (2) design, validation, and administration of a structured questionnaire, (3) statistical analysis of responses. In the first stage, a systematic literature review of 27 peer-reviewed studies was conducted to extract a list of managerial competencies relevant to construction project settings. The second stage involved developing a Likert-scale questionnaire, which was calibrated through expert judgment and administered to two distinct groups: construction managers and students enrolled in construction-related programs. Participants rated the perceived usefulness of AI in enhancing each of the identified competencies. The third stage included a statistical analysis of the survey responses to determine which competencies are most likely to be strengthened through AI adoption. A qualitative comparison of the perspectives of students and professionals further enriched this analysis. Fig. 1 presents a visual summary of each stage of the research method adopted.

This study adopts a post-positivist and pragmatic research philosophy, acknowledging that while reality exists independently, it can only be partially understood through observation, interpretation, and statistical inference. Within this paradigm, knowledge is developed through empirical evidence and systematic analysis, allowing generalizable insights while recognizing contextual limitations. The post-positivist stance aligns with the study's objective of quantifying perceptions of AI influence on managerial competencies, as it relies on measurable indicators obtained through structured instruments [86]. At the same time, a pragmatic orientation guides the methodological design by emphasizing usefulness and practical implications for the construction industry and educational settings. This philosophical foundation supports the combination of a systematic literature review and a quantitative survey design, followed by inferential analyses such as RII, Mann–Whitney U, and EFA, which seek to test and refine understanding through observable evidence.

3.1. Identification of competencies to manage construction projects

The identification of key competencies for effective construction project management was conducted through a structured and systematic literature review, based on the methodologies reported by Castañeda et al. [87], Herrera et al. [88] and Umar [89]. The process started with formulating a guiding research question: "What are the core competencies necessary for managing construction projects?" This question served as the foundation for developing the search strategy. As a result, a Boolean search string was constructed, which included relevant terms such as ["competency" OR "skill"] AND ("construction") AND ("project management" OR "management" OR "administration") AND ("leadership" OR "communication" OR "planning" OR "coordination") AND ("risk" OR "quality" OR "time" OR "cost"), ensuring comprehensive coverage of the topic. To guarantee the reliability and scientific quality of the sources, the Scopus database was chosen due to its reputable indexing of peer-reviewed literature in engineering, management, and applied technologies. The initial search produced 954 documents. After applying filters for publication years (2014–2025) and language (English), the dataset was narrowed down to 549 articles. Further refinement by document type (articles only) and subject areas ("Engineering," "Computer Science," and "Business, Management, and Accounting") resulted in a final set of 206 relevant publications. A three-tier inclusion/exclusion criterion was then applied to ensure the relevance of the selected studies. Studies were included if they: (1) focused explicitly on construction projects, (2) addressed the topic of professional competencies within the construction industry, and (3) reported managerial competencies specific to construction project managers.

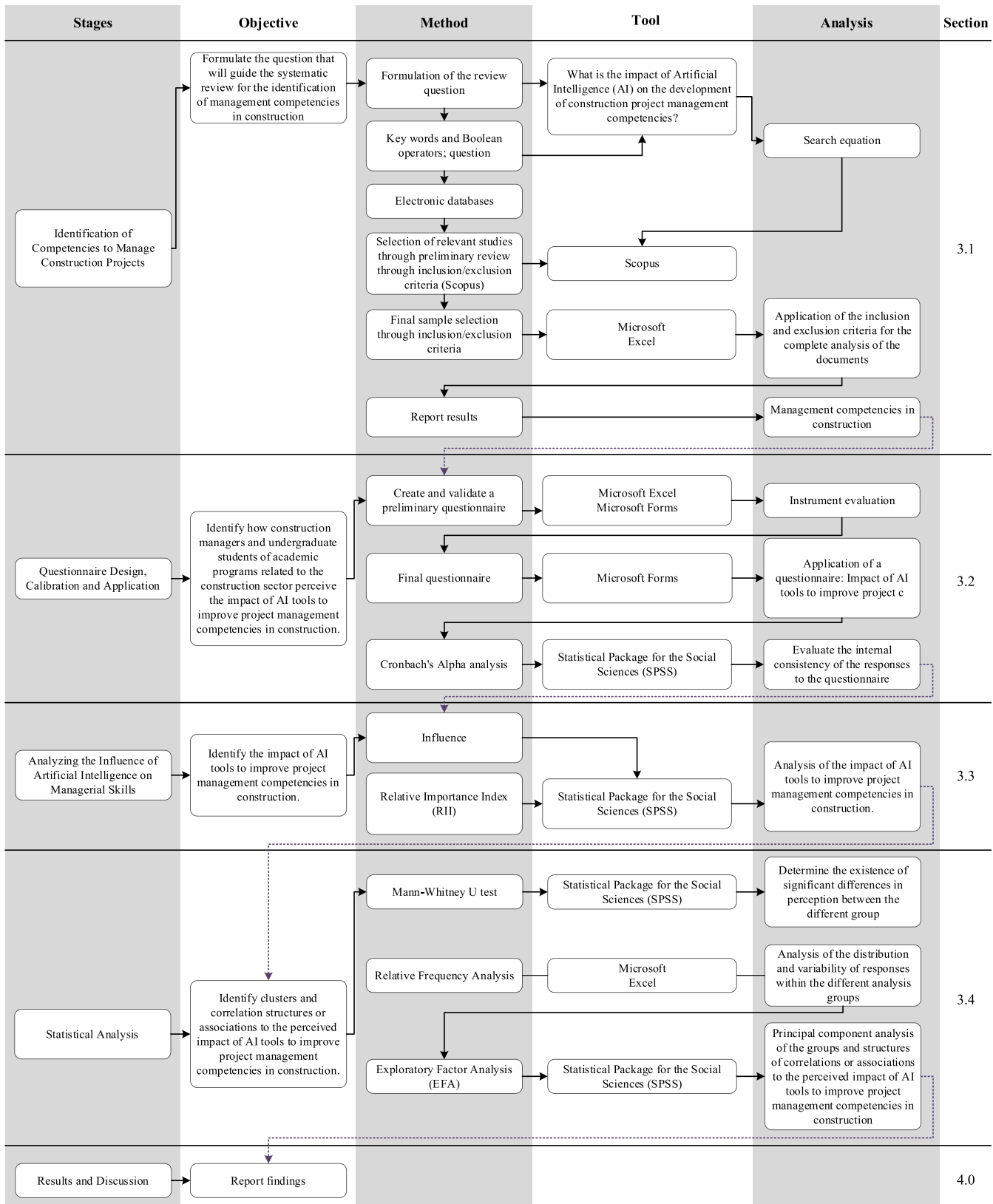


Fig. 1. . Characteristics of the research method stages.

Based on the analysis of 206 selected documents, a systematic extraction of managerial competencies relevant to construction project management was conducted in three distinct phases: (1) initial extraction, during which 35 competencies were identified; (2) categorization

and consolidation, where the competencies were grouped into seven thematic categories according to their functional characteristics and relevance to construction project contexts; and (3) expert validation. In this final phase, a panel of specialists in project management and

emerging technologies reviewed the compiled list (see Table 1), eliminating redundancies, resolving overlaps, and confirming the practical relevance of each item. As a result, a final set of 35 core competencies was defined and subsequently used as the foundation for the next stage of the study, which aimed to assess the perceived potential of AI-based tools to support the development of these competencies among both construction professionals and students.

3.2. Questionnaire design and application

A structured survey instrument was designed to explore the views of students enrolled in construction management courses, as well as those of practicing construction managers, on the influence of artificial intelligence (AI) tools in fostering key competencies for managing construction projects. The questionnaire consisted of three main parts. The initial section focused on informed consent, detailing the ethical considerations regarding data use, confidentiality, and the protection of participant information. It also provided a brief description of the study's purpose and scope. Participants were asked to indicate their agreement to participate; an affirmative response allowed them to proceed to the next section, while a negative response triggered a thank-you message and ended the survey. The second section gathered key categorical variables to contextualize the respondents' profiles. These variables included years of professional experience, roles within the construction sector, types of projects participants were engaged in, and the countries where they gained their experience. For student respondents, additional questions were asked to capture their academic program and current level of study. The third section focused on evaluating the 35 competencies identified in the literature review. Participants were asked to rate how beneficial they perceived AI-based tools (such as ChatGPT, Gemini Copilot, or DeepSeek) to be in enhancing each of the competencies. The questionnaire employed a five-point Likert scale, with response options ranging from 1 ("No influence") to 5 ("Very high influence"). This scaling method enabled a quantitative evaluation of how students and professionals perceive the ability of intelligent digital assistants to support the development of competencies in construction project management.

The respondent profile consisted of two target populations: experienced professionals in construction project management and advanced students in construction management and applied technologies (see Fig. 2). To ensure the validity and reliability of the questionnaire, the instrument was reviewed by a panel of six experts, each with at least five years of experience in construction project management and knowledge of AI-powered conversational assistants such as ChatGPT, Copilot, Gemini, or DeepSeek. These experts provided feedback that helped refine the clarity, relevance, and structural coherence of the items, resulting in improvements to the overall design of the instrument. After incorporating the expert feedback, the final version of the questionnaire was distributed online to a purposive sample of participants. Construction managers were selected based on their professional roles and active involvement in project delivery, while students were chosen according to their academic programs and levels of progression. This selection strategy ensured representation from both industry practitioners and emerging professionals in the training program. A total of 150 questionnaires were distributed, and 135 completed responses were

received, resulting in a response rate of 90 %. This rate was considered sufficient to support robust statistical analysis in the subsequent phase of the study.

While the survey achieved a satisfactory overall sample size ($n = 135$), the proportion between students ($n = 82$) and professionals ($n = 53$) was not evenly balanced. This asymmetry may have slightly influenced the results in favor of the academic perspective. Nevertheless, the sampling strategy was purposive and intentionally designed to contrast perceptions between two distinct stakeholder groups: emerging professionals under academic training and experienced construction managers actively engaged in project delivery. Similar to the approach adopted by Umar et al. [90] considered purposive sampling for exploratory studies aimed at identifying perceptual gaps rather than achieving population representativeness. This study acknowledges the uneven distribution between groups as a methodological limitation; however, it also recognizes that this composition aligns with the research objective of comparing how exposure to artificial intelligence differs across educational and professional contexts.

A reliability analysis was conducted using Cronbach's alpha coefficient to ensure the internal consistency of the measurement instrument. This statistical approach assesses the consistency with which items within a Likert-type scale measure the same underlying construct, thereby reflecting the internal coherence of participants' responses. Based on the criteria proposed by George and Mallery [91], Cronbach's alpha values are categorized as follows: $\alpha \geq 0.9$ (excellent), $0.8 \leq \alpha < 0.9$ (good), $0.7 \leq \alpha < 0.8$ (acceptable), $0.6 \leq \alpha < 0.7$ (questionable), $0.5 \leq \alpha < 0.6$ (poor), and $\alpha < 0.5$ (unacceptable). The reliability assessment in this study produced an alpha coefficient of 0.89, demonstrating strong internal consistency. This result confirms the robustness of the instrument and supports the reliability of the responses obtained from both student and professional participants within the construction sector.

The instrument was administered to two specific target populations: practicing construction managers and advanced students in the field of construction management. The authors used purposive sampling with the explicit goal of contrasting perceptions between professionals in the field and those in training. The authors recognize that this approach excludes other stakeholder groups in the sector, which limits the overall representativeness of our findings. However, this decision aligns with the exploratory nature of the study and is supported by the research agenda discussed later. As noted in this section, the distribution of participants across subgroups was uneven, with 82 students and 53 professionals. This imbalance represents an additional methodological limitation that may slightly bias the results in favor of the academic perspective.

3.3. Analyzing the influence of artificial intelligence on managerial skills

The evaluation of scores from construction students and managers started with the calculation of the Relative Influence Index (RII) (Eq. (1)). This was done to rank the 35 managerial competencies relevant to construction projects based on how much they could be enhanced through the use of AI-enabled conversational assistants, such as ChatGPT, DeepSeek, Copilot, and Gemini. The RII transforms the ordinal data collected using a five-point Likert scale (where 1 = "No influence" and 5 = "Very high influence") into a continuous metric that ranges from 0 to 1. This transformation enables direct comparisons across different groups and minimizes subjective judgment noise. A total of 82 students enrolled in construction-related programs and construction management courses, along with 53 practicing construction managers, participated in the survey. Their demographic and professional characteristics are summarized in Fig. 2. Before calculating the RII, the internal consistency of the instrument was confirmed (Cronbach's $\alpha = 0.91$), and Harman's single-factor test showed no significant common-method bias [92]. The resulting indices create a clear hierarchy of competencies and quantify the level of agreement regarding the potential impact of AI. This highlights the training areas with the most significant opportunities

Table 1
Role and position of professionals.

Id	Role	Position	Profession	Years of experience
P ₁	Consultant	Project manager	Civil Engineer, MSc, PhD	19
P ₂	Contractor	Project manager	Civil Engineer, MSc	15
P ₃	Contractor	Company manager	Civil Engineer, MSc	12

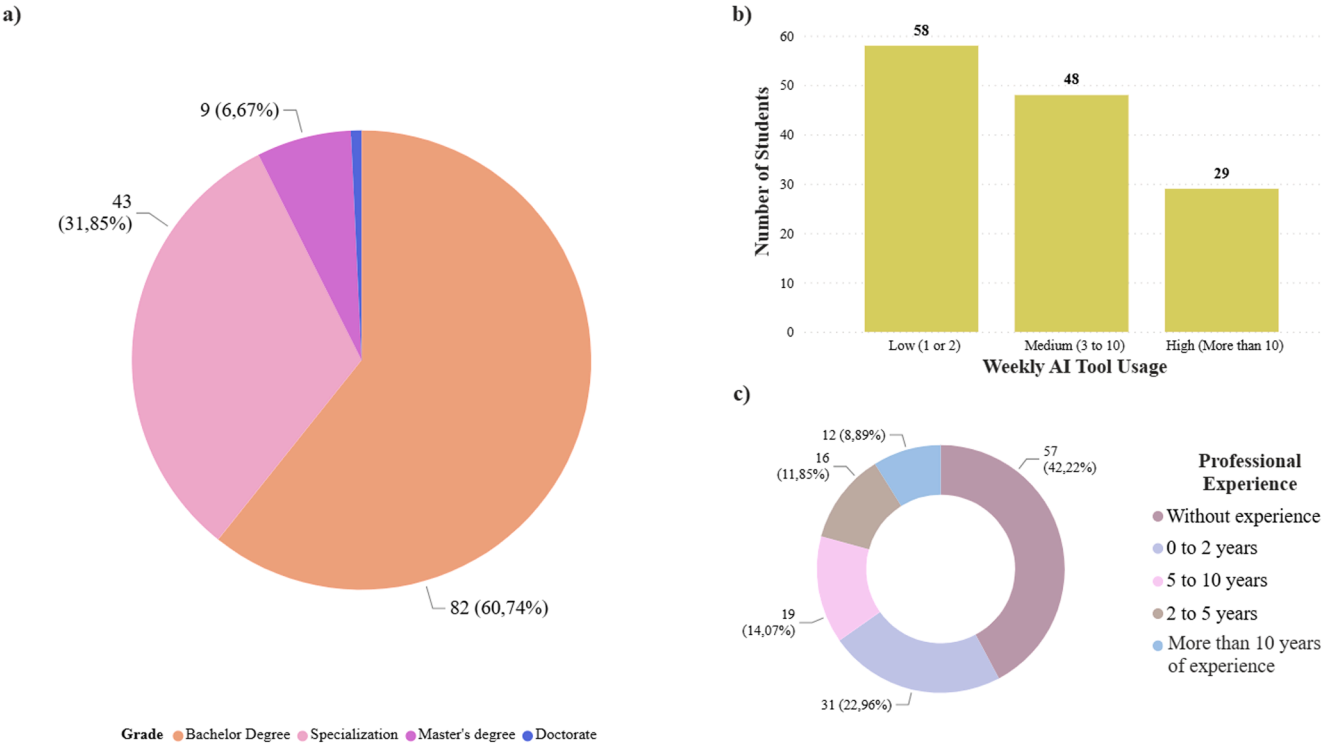


Fig. 2. . Profile characteristics of questionnaire participants.

for improvement through emerging technologies.

$$RII = \left(\frac{\sum_{i=1}^N w_i}{N * w_{max}} \right) \tag{1}$$

Where w_i is the classification weights, assigned by the student or manager i ; N is the total number of valid responses for the questionnaire ($N = 135$); and w_{max} is the maximum possible level of influence for classification ($w_{max} = 5$), based on the evaluation scale in the questionnaire.

3.4. Statistical analysis of responses

The Mann–Whitney U test was used to determine whether the two stakeholder groups (construction managers and construction students) differed significantly in their assessment of AI-enabled competency enhancement. This non-parametric procedure is particularly suited for independent samples with underlying distributions that deviate from normality, which is common with ordinal Likert-type data. Since the test is rank-based, it is robust against both non-normality and heteroscedasticity. In the present study, the Mann-Whitney U test was conducted for each of the eleven competencies previously identified, comparing the perceptions of students and practicing professionals regarding the extent to which AI tools can enhance those capabilities. Table 2 summarizes the analytical groups and the distributional characteristics that justified the use of the Mann-Whitney U test. The resulting U statistics and associated p-values were interpreted in conjunction with group medians and effect-size estimates. By triangulating these statistical findings with relative frequency patterns, this study identified competencies that, while generally regarded as important, show significant divergence between early-career and seasoned respondents. These differences carry specific implications for targeted training interventions and technology adoption roadmaps in the construction sector.

To further refine these findings, the analysis also considered whether experience level and role type influence perceptions of AI’s impact on

Table 2
. Analysis groups for the Mann–Whitney U test.

Variables of analysis	Categorical variables	Groups of analysis
Experience	Without experience 0 to 2 years 2 to 5 years 5 to 10 years >10 years	Without experience With experience
Use	Low (1 or 2 times per week) Medium (3 to 10 times per week) High (>10 times per week)	Low use Medium and high use
Role	Students in AI-based construction programs Construction managers	Students Construction managers

managerial competencies. Participants were grouped by years of professional experience (0–2, 2–5, 5–10, and >10 years) and by role (student vs. practicing manager). This approach allowed for the identification of differences in perceptions of how AI tools affect specific competencies across various experience levels and professional roles.

In addition to these non-parametric tests, an exploratory factor analysis (EFA) was conducted to reduce a large set of correlated competencies into a smaller group of statistically independent factors that maximize the explained variance in the dataset. Before extraction, two diagnostics confirmed that the sample was adequate and that factor analysis was suitable for the data. The Kaiser–Meyer–Olkin (KMO) index yielded a value of 0.928, which is considered "marvellous" according to Kaiser’s original scale [93]. Additionally, Bartlett’s test of sphericity was highly significant ($p < 0.001$), indicating sufficient inter-item correlations. These results justified the use of principal-component extraction and varimax rotation. The resulting solution distilled the initial competency list into N latent dimensions, which collectively accounted for 70.2 % of the observed variance, revealing the underlying conceptual framework of construction management skills. By grouping

competencies with similar response patterns, the analysis identified coherent skill clusters that can be simultaneously enhanced through AI-enabled interventions. This insight can inform both curriculum design and professional development strategies within the construction sector.

4. Results

This chapter is organized into four analytical sections, each aligned with the research objectives of the study. [Section 4.1](#): Construction Management Competencies introduces the 35 key managerial competencies identified through a methodological process that involved a literature review and survey responses. [Section 4.2](#): Perceptions of Artificial Intelligence's Influence on Managerial Competencies examines respondents' views on the impact of AI-based tools on the identified competencies. Utilizing the Relative Influence Index (RII), calculated from responses provided by 135 participants, this section highlights which competencies are most significantly affected by the adoption of AI in construction project management. [Section 4.3](#): AI-Influenced Competency Domains for Construction Management presents findings from an exploratory factor analysis, which statistically derived a set of competency domains significantly influenced by the integration of AI. These domains show how AI technologies can help managerial roles and skill requirements in the sector. Finally, [Section 4.4](#): Comparison of Perceptions between Analysis Groups investigates the statistically significant differences in perceptions between students and construction managers, using the Mann-Whitney U test.

4.1. Construction management competences

[Table 3](#) presents the key competencies necessary for effective construction project management, identified through a systematic literature review to determine the essential capabilities a manager must possess in this field. The findings reveal a broad and diverse range of competencies, categorized into two main groups: soft skills and technical skills. These competencies include areas such as scheduling, planning, budgeting, and decision-making in complex and uncertain environments. To facilitate analysis, the competencies have been organized into seven categories: (1) leadership and people management, (2) project management, (3) technical competencies, (4) safety and regulatory compliance, (5) communication and interpersonal skills, (6) strategic thinking and analytical skills, and (7) personal development and professional ethics. This classification shows an overview of the competency profile required to lead construction projects, integrating both interpersonal and technical dimensions. Furthermore, the results indicate that several of these competencies, particularly those related to data analysis, strategic planning, and operational control, can be improved by adopting artificial intelligence (AI) tools. These emerging technologies have the potential to automate repetitive tasks, uncover complex patterns within large datasets, and deliver process information that supports more accurate and informed decision-making throughout the project lifecycle.

4.2. Perceptions of artificial intelligence influence on managerial competences

[Table 4](#) displays the Relative Influence Index (RII) for each competency assessed in this study. These indices are derived from responses collected from two distinct groups: 82 students enrolled in construction management programs that incorporate artificial intelligence (AI) tools into their curricula, and 53 construction managers who actively use AI-based tools to support their project management tasks. The analysis reveals some differences in perceptions between students and professionals, particularly regarding the practical impact of AI on some management skills. However, there is significant agreement on the overall recognition of AI's potential to enhance key managerial competencies in the construction sector. This shared perspective reveals a

Table 3
Construction management competences.

Category	Id	Competence	Description	References
Leadership and People Management	C ₁	Leadership	Guide and motivate the team to achieve project objectives.	[46,47,94–107]
	C ₂	Teamwork	Promote collaboration among architects, engineers, and contractors.	[47,54,57,94–98,100,103,105,106,108–111]
	C ₃	Effective Delegation	Assign appropriate tasks to team members based on their skills.	[54,94,95,97,98,105,109,110]
	C ₄	Conflict Management	Resolve disputes between workers or contractors in a fair manner.	[54,57,94,99–101,105,106,109,112]
	C ₅	Mentoring	Train and develop skills in personnel.	[97,102,103]
Project Management	C ₆	Planning	Develop detailed work plans with defined timelines and resources.	[46,47,51,54,57,95,97,100–102,105,109,111,113]
	C ₇	Cost Control	Monitor and adjust the budget to avoid cost overruns.	[46,47,51,54,57,94,95,99–101,103–107,109,113,114]
	C ₈	Risk Management	Identify and mitigate risks that may affect the project.	[46,54,57,94,95,100,101,103–107,109,113,114]
	C ₉	Change Management	Adapt the project in response to unexpected modifications.	[57,95,103,114]
	C ₁₀	Quality Control	Ensure materials and processes meet required standards.	[46,47,54,57,94,95,99–101,103,105,106,109,113,114]
Technical Competencies	C ₁₁	Blueprint Interpretation	Read and understand blueprints from various project areas.	[57,100]
	C ₁₂	BIM Utilization	Operate Building Information Modeling tools.	[54,98]
	C ₁₃	Material Knowledge	Select appropriate materials for each phase of the project.	[94,100,114]
	C ₁₄	Digital Tools Proficiency	Use specialized software (e.g., AutoCAD, MS Project, BIM).	[46,47,54,57,94,95,99,102,114]
	C ₁₅	Technological Innovation	Implement new technologies in project development.	[54,98,100,101,103,109]
Safety and Regulatory Compliance	C ₁₆	Occupational Safety	Ensure a safe working environment for all personnel.	[47,57,94,95,100,101,114,115]

(continued on next page)

Table 3 (continued)

Category	Id	Competence	Description	References
Communication and Interpersonal Skills	C ₁₇	Regulatory Compliance	Ensure the project complies with local laws and regulations.	[57,100,109]
	C ₁₈	Environmental Management	Minimize the project's environmental impact (e.g., waste reduction).	[57,95,100,105,106,114]
	C ₁₉	Document Control	Organize and manage permits, contracts, and warranties.	[57,95,100,105,106,114]
	C ₂₀	Site Audits	Conduct inspections to verify compliance with standards.	[116]
	C ₂₁	Effective Communication	Convey clear information to clients, teams, and suppliers.	[46,47,51,54,57,94,95,97–101,103–107,109–113]
	C ₂₂	Negotiation	Reach mutually beneficial agreements with suppliers and contractors.	[54,94,95,97,100,101,109,112]
	C ₂₃	Supplier Management	Coordinate the delivery of materials and services.	[47,54,95,99,105,106,113]
Strategic Thinking and Analytical Skills	C ₂₄	Customer Service	Respond to client needs and expectations.	[47,96,99,100]
	C ₂₅	Reporting	Prepare clear reports on project progress.	[100,116]
	C ₂₆	Decision-Making	Evaluate alternatives and choose the best option for the project.	[46,54,94,97–99,101,103,105,107–109,112]
	C ₂₇	Critical Thinking	Analyze data and situations to make informed decisions.	[46,54,57,95,96,103,107–111]
	C ₂₈	Strategic Thinking	Plan long-term strategies for project or business growth.	[47,54,95,99,100,102,104,107–110]
Personal Development and Professional Ethics	C ₂₉	Supply Chain Management	Ensure timely procurement of materials.	[54,57,98]
	C ₃₀	Resource Optimization	Maximize efficiency in the use of labor and equipment.	[47,54,57,99,100,102,105,109,110]
	C ₃₁	Continuous Learning	Stay up-to-date with new construction technologies and methods.	[47,54,57,100,102,107,109]
	C ₃₂	Adaptability	Adjust to changes in project scope or external conditions.	[54,57,96,97,100,101,103,107,109]
	C ₃₃	Professional Ethics	Act with honesty and transparency in all decisions.	[46,54,94,97,100,101,103,105,107]
	C ₃₄	Resilience	Remain calm and productive in crisis situations.	[96,97,108]
	C ₃₅	Results Orientation	Focus on efficiently	[100,110]

Table 3 (continued)

Category	Id	Competence	Description	References
			achieving project objectives.	

growing consensus on the strategic role that intelligent algorithms and digital assistants play in enhancing the skill sets necessary for effective construction project management.

Technological Innovation (C₁₅) has been identified as the managerial competence most enhanced by AI adoption in construction project management, boasting the highest score (RII = 0.846). This finding reflects a broad recognition of AI's role in enhancing analytical capacity and supporting decision-making across project workflows. Generative platforms such as ChatGPT, DeepSeek, Copilot, and Gemini transform dispersed data into actionable insights, supporting process improvement and the development of technical solutions throughout the project life-cycle. During the conceptual design phase, these tools generate alternative layouts that optimize costs, energy consumption, and carbon emissions. In the planning phase, they automatically create schedules and resource plans based on historical data. On-site, they analyze sensor data to identify deviations and propose adaptive sequencing. Finally, during project close-out, they enhance digital twin models with lessons learned to facilitate continuous organizational learning. By automating routine synthesis and reducing the cognitive load associated with research, documentation, and ideation, these tools enable teams to focus on higher-value tasks, promote faster iteration, strengthen strategic decision-making, and foster cross-disciplinary collaboration [117].

Planning (C₆) ranked second (RII = 0.834), highlighting the important role of AI in improving schedule accuracy and resource allocation in construction projects. Modern AI platforms enhance traditional predictive analytics by offering conversational interfaces that translate work breakdown structures into logically linked Gantt charts, facilitating a more intuitive and streamlined workflow. They also enable real-time simulations to anticipate potential delays, quantify their time and cost implications, and generate adaptive alternatives to maintain schedule performance [118]. These generative capabilities reduce the time between field data collection and program updates, enabling managers to adjust schedules and meet contractual milestones iteratively. AI-assisted planning has redefined the professional profile of project managers, who now require stronger analytical reasoning, data literacy, and the ability to validate automated results. However, the effectiveness of these tools depends on the availability of high-quality and consistent underlying data [119]. As a result, AI adoption in planning has transformed the role of project managers, requiring them to employ analytical reasoning, demonstrate fluency in AI-generated models, and increase their awareness of ethical considerations. To fully realize the potential of AI-enabled planning, universities and professional organizations must integrate data-driven decision-making, prompt engineering, and ethical AI into their curricula and continuing education programs.

Reporting (C₂₅) has been identified as the third most strengthened managerial competence through AI adoption (RII = 0.831), reflecting its growing role in automating progress tracking and performance communication in construction projects. Traditionally, generating reports has been a time-consuming and error-prone manual task, often resulting in inconsistencies and delays in decision-making. AI tools are transforming this process by automating the collection, processing, and synthesis of data from various sources, including sensors, Building Information Modeling (BIM) systems, construction management tools, and digital twins [41]. This automation produces real-time reports that enhance traceability, transparency, and responsiveness. Generative systems assist managers in drafting structured summaries, translating large datasets into concise insights, detecting anomalies, and adapting content for specific audiences such as technical teams, clients, or regulators. These capabilities accelerate information flow and increase the strategic value of reporting by supporting faster, evidence-based

Table 4

. Relative Influence Indices of AI on construction project management competencies.

Id	Competencies to manage construction projects	Students		Managers		Overall	
		RII	Rank	RII	Rank	RII	Rank
C ₁₅	Technological Innovation: Implement new technologies in project development	0.820	1	0.758	1	0.846	1
C ₆	Planning: Develop detailed work plans with defined timelines and resources	0.820	1	0.698	4	0.834	2
C ₂₅	Reporting: Prepare clear reports on project progress.	0.807	3	0.725	2	0.831	3
C ₁₉	Document Control: Organize and manage permits, contracts, and warranties	0.783	6	0.675	9	0.803	4
C ₇	Cost Control: Monitor and adjust the budget to avoid cost overruns	0.795	5	0.675	9	0.803	4
C ₃₁	Continuous Learning: Stay up-to-date with new construction technologies and methods	0.776	7	0.694	5	0.800	6
C ₁₂	BIM Utilization: Operate Building Information Modeling tools	0.771	9	0.691	6	0.797	7
C ₈	Risk Management: Identify and mitigate risks that may affect the project	0.771	9	0.713	3	0.797	7
C ₁₃	Material Knowledge: Select appropriate materials for each phase of the project	0.763	13	0.672	12	0.787	9
C ₃₀	Resource Optimization: Maximize efficiency in the use of labor and equipment	0.798	4	0.657	15	0.782	10
C ₁₇	Regulatory Compliance: Ensure the project complies with local laws and regulations	0.771	9	0.683	7	0.782	10
C ₂₈	Strategic Thinking: Plan long-term strategies for project or business growth	0.759	14	0.679	8	0.779	12
C ₅	Mentoring: Train and develop skills in personnel	0.766	12	0.664	14	0.778	13
C ₃	Effective Delegation: Assign appropriate tasks to team members based on their skills	0.776	7	0.638	19	0.772	14
C ₁₄	Digital Tools Proficiency: Use specialized software (e.g., AutoCAD, MS Project, BIM)	0.732	22	0.675	9	0.767	15
C ₃₂	Adaptability: Adjust to changes in project scope or external conditions	0.746	17	0.638	19	0.761	16
C ₉	Change Management: Adapt the project in response to unexpected modifications	0.737	21	0.668	13	0.761	16
C ₂₆	Decision-Making: Evaluate alternatives and choose the best option for the project	0.741	20	0.645	17	0.753	18
C ₂₇	Critical Thinking: Analyse data and situations to make informed decisions	0.732	22	0.657	15	0.750	19
C ₃₅	Results Orientation: Focus on efficiently achieving project objectives	0.751	16	0.642	18	0.748	20
C ₂₄	Customer Service: Respond to client needs and expectations	0.746	17	0.634	21	0.744	21
C ₂₁	Effective Communication: Convey clear information to clients, teams, and suppliers	0.759	14	0.604	25	0.739	22
C ₂₉	Supply Chain Management: Ensure timely procurement of materials	0.715	24	0.634	21	0.730	23
C ₁₀	Quality Control: Ensure materials and processes meet required standards	0.710	26	0.626	23	0.723	24
C ₁₈	Environmental Management: Minimize the project's environmental impact (e.g., waste reduction)	0.746	17	0.574	29	0.719	25
C ₁₁	Blueprint Interpretation: Read and understand blueprints from various project areas	0.695	28	0.600	26	0.713	26
C ₂₃	Supplier Management: Coordinate the delivery of materials and services	0.693	29	0.615	24	0.701	27
C ₁₆	Occupational Safety: Ensure a safe working environment for all personnel	0.712	25	0.532	31	0.690	28
C ₂₀	Site Audits: Conduct inspections to verify compliance with standards	0.685	31	0.577	27	0.674	29
C ₂₂	Negotiation: Reach mutually beneficial agreements with suppliers and contractors	0.698	27	0.525	33	0.671	30
C ₂	Teamwork: Promote collaboration among architects, engineers, and contractors	0.688	30	0.562	30	0.671	30
C ₃₄	Resilience: Remain calm and productive in crisis situations	0.663	35	0.577	27	0.670	32
C ₄	Conflict Management: Resolve disputes between workers or contractors in a fair manner	0.685	31	0.532	31	0.661	33
C ₁	Leadership: Guide and motivate the team to achieve project objectives	0.678	33	0.509	35	0.652	34
C ₃₃	Professional Ethics: Act with honesty and transparency in all decisions	0.671	34	0.513	34	0.644	35

decisions throughout project execution. Effective implementation, however, depends on integration across technological systems, data standardization, and users' ability to interpret AI-generated outputs critically. In collaborative settings, automated reporting enhances coordination and trust among stakeholders by providing consistent and verifiable information almost instantaneously.

Document Control (C₁₉) was identified as the fourth managerial competence most impacted by AI tools (RII = 0.803). This highlights a widespread acknowledgment of the benefits of AI in efficiently managing project documentation. Construction projects often produce vast amounts of technical, contractual, and administrative information that need to be organized, updated, and made easily accessible. AI applications streamline this process by automating classification, version tracking, and error detection, thereby reducing retrieval time and improving compliance. Advanced systems also issue alerts on missing documentation, upcoming deadlines, or unfulfilled regulatory requirements, features that are particularly valuable in large or multi-contract projects [120]. AI tools further improve document control by assisting managers in drafting structured reports throughout various project phases, including design reviews, scheduling summaries, and operational updates. These systems can translate technical content into clear narratives and extract key action points from unstructured data. They also function as intelligent search engines, capable of retrieving and summarizing relevant documents in seconds. However, for AI to be effective in document control, it is crucial to maintain data integrity, establish access protocols, and ensure user accountability. Interoperability across digital platforms is also crucial to prevent data silos, duplication, and information loss. Establishing strong document governance policies and fostering an organizational culture centered on traceability and information transparency are vital for solidifying AI's

role in this core function of construction project management.

Cost Control (C₇) has been identified as the fifth most significant managerial competence influenced by AI (RII = 0.803). This highlights its critical role in maintaining financial discipline within projects. AI systems can provide real-time expense monitoring and continuously compare expenditures against baseline budgets, allowing for the prompt detection of deviations and the proposal of corrective actions before cost overruns occur. This capability is particularly beneficial in volatile environments where material prices, labor availability, and subcontractor performance can change rapidly [121]. Additionally, AI tools facilitate sensitivity analyses and scenario-based simulations, aiding in strategic financial forecasting throughout the design, planning, and operational phases. Generative AI platforms enhance this process by enabling project managers to efficiently produce budget reports, draft variance justifications, synthesize cash flow projections, and even recommend procurement alternatives based on market insights. These systems can also identify anomalies in financial data through natural language queries, reducing the need for complex manual spreadsheet audits. To fully capitalize on these advantages, it is vital to ensure access to reliable and up-to-date accounting data, along with trained teams capable of interpreting alerts and taking proactive measures. The implementation of interactive dashboards powered by AI enhances the visibility and accessibility of key financial indicators, promoting a more agile and results-oriented management approach. Additionally, integrating digital finance competencies into engineering and architecture education will be crucial for preparing a new generation of professionals who can effectively use AI as a core tool in cost control for construction projects.

The managerial competencies with the greatest perceived influence of artificial intelligence, as presented in Table 4, Technological Innovation (C₁₅), Planning (C₆), Reporting (C₂₅), and Document Control

(C₁₉), provide a solid foundation for deriving operational implications aimed at industrial adoption. In the training domain, it is recommended to institutionalize short and recurring programs on data literacy and responsible AI use, complemented by specialized training in prompt design for the AEC context and in the integration of BIM and AI workflows focused on semantic search, schedule assistance, and document intelligence. These initiatives should emphasize the critical interpretation of AI outputs, systematic validation of results, and the inclusion of human oversight mechanisms in high-impact or safety-critical tasks. At the organizational level, it is advisable to appoint AI focal points at the project level, clearly define roles for data stewardship and quality assurance, and implement model and prompt lifecycle management practices that ensure version control, traceability, and periodic performance evaluation. Regarding governance and compliance, internal policies should specify data standards, audit records, and bias-assessment procedures, while procurement processes should include transparency and security requirements for AI-enabled solutions.

Building on these operational and organizational recommendations, it is also important to consider how perceptions of AI influence differ across managerial responders. While technical and analytical skills appear to benefit most from AI integration, as highlighted earlier, not all competencies share this insight. As shown in Table 4, practicing construction managers assigned lower RII scores to competencies such as leadership (C₁; RII = 0.509) and professional ethics (C₃₃; RII = 0.513), compared to other technical and analytical skills. This result differs from student respondents, who rated these competencies higher (leadership: RII = 0.678; professional ethics: RII = 0.671). This suggests that managers perceive AI tools as less impactful for competencies rooted in interpersonal relationships, situational judgment, and ethical decision-making. Several factors may contribute to this perception. First, leadership and ethics in construction are traditionally developed through direct human interaction, mentorship, and organizational culture. Which are areas under limited AI influence. Second, managers have practical constraints such as resistance and regulatory requirements that may further limit the perceived importance of AI for these competencies.

4.3. AI-influenced competency domains for construction management

An assessment of how the incorporation of Artificial Intelligence (AI) tools enhances management competencies in construction projects was conducted through an Exploratory Factor Analysis (EFA), using the responses provided by participants, as presented in Section 3.2. This analysis aims to reduce the dimensionality of the initial set of 35 managerial competencies by identifying correlations and groupings, resulting in a total of six components with eigenvalues greater than one and robust factor loadings of greater than 0.2. Each component was subsequently interpreted based on the items with the highest weights; this led to the definition of six thematic categories that structure the discussion: (1) leadership and collaborative management, (2) adaptability and results orientation, (3) project control and risk management, (4) strategic decision and communication, (5) technical and digital skills, and (6) compliance and environmental management (see Appendix A). These six dimensions capture most of the explained variance and help to understand how AI-based solutions contribute to the development of managerial skills to lead construction works, optimize resources, and ensure project sustainability.

The integration of these AI findings into real practice is essential, and doing so requires changes at the university, industry, and policy-making levels. Institutions can embed competencies into their curriculum through modules on ethics, data analytics, and prompt engineering. Incorporating hands-on, problem-based projects, using real construction datasets, and using AI tools for analysis (e.g., predictive modeling, generative design). The construction industry can facilitate the adoption of AI-driven platforms for project management activities such as project scheduling, cost estimating, and risk analysis. Invest in workforce development programs focusing on AI literacy and data interpretation.

Additionally, establish guidelines for the ethical use of AI. Similarly, policymakers can develop a framework by promoting standardization in the recognition of AI skills, offering incentives for the use of AI, promoting AI development, and ensuring the ethical deployment of AI.

AI platforms, which integrate predictive analytics, computer vision, and natural language processing, among other technologies, have emerged as promising tools in the construction sector by enhancing the performance of construction management competencies. These tools generate a continuous flow of information that strengthens the nine competencies grouped under the Leadership and Collaborative Management category. Platforms such as Procore Copilot centralize contractual documents, drawings, technical specifications, and performance metrics, while incorporating intelligent assistants that automate communication drafting, contract analysis, and report generation [122, 123], thereby reinforcing competencies such as Leadership (0.83) and Professional Ethics (0.64). By facilitating smooth coordination among various project stakeholders, these platforms also contribute directly to the development of Teamwork (0.78), promoting more transparent communication and effective decision traceability [124]. In parallel, tools like Buildots, which use computer vision through mobile cameras or fixed sensors, enable automated on-site inspections to detect construction deviations or unsafe conditions, supporting competencies such as Occupational Safety (0.58) and Site Audits (0.56) by generating immediate alerts regarding the absence of PPE or operational risks [125, 126]. ALICE Technologies, through artificial intelligence algorithms applied to construction scheduling, optimizes activity sequences, reduces uncertainty, and generates viable alternatives to address time and resource constraints [127,128], thereby strengthening competencies related to Negotiation (0.59) and Conflict Management (0.56) by enabling better preparation for contractual discussions and conflict resolution. Finally, the incorporation of predictive models in logistics processes enhances Supplier Management (0.52) and Supply Chain Management (0.37) by anticipating disruptions, ranking suppliers based on performance, and optimizing supply routes, thereby mitigating tensions arising from delays or cost overruns [129].

The second domain, Adaptability and Results Orientation, encompasses a set of interrelated competencies that enable the construction manager to sustain performance in volatile environments. Adaptability (0.67) and Resilience (0.59) form the reactive core through which plans are adjusted in response to change. Operating upon this core are Results Orientation (0.45) and Resource Optimization (0.47), which transform flexibility into concrete time-cost goals and efficient use of labor and equipment. The human dimension is addressed by Mentoring (0.67) and Customer Service (0.65), as training the team and aligning client expectations facilitate the acceptance and implementation of scope changes. Finally, Continuous Learning (0.63) acts as a bridge that injects updated knowledge (such as standards, technologies, and methods) into the rest of the competencies, ensuring that each adaptation leads to performance improvement rather than mere change. AI-based tools can simulate alternative scenarios and recommend the one that maximizes outcomes with the available resources, such as by detecting on-site deviations in comparison with the BIM model and generating early alerts that both demand and support adaptive responses. These tools can also automate data extraction from RFIs and submittals, among other functions [129,130].

The third domain, Project Control and Risk Management, functions as a continuous regulatory circuit that preserves the time-cost-quality triangle throughout the construction project life cycle. Risk Management (0.76) focuses on monitoring internal and external sources of uncertainty; each identified threat activates Change Management (0.65) protocols that redefine scope, resources, or deadlines before the impact materializes. These decisions are supported by Planning (0.65), which incorporates milestones, buffer zones, and alternative paths, ensuring that adjustments can be integrated without compromising Quality Control (0.69), which is maintained through inspections and normative benchmarks. Cost Control (0.54) compares actual and projected

expenditures, triggers deviation alerts, and guides the reallocation of resources, while Reporting (0.47) consolidates progress metrics, residual risks, and financial performance into executive dashboards that support informed and timely decision-making. Early detection drives adaptation, which in turn relies on robust planning, and outcomes are validated through quality assurance, budget compliance, and transparent communication [131,132]. AI tools enable scenario simulation and contingency analysis to propose alternative schedules with reduced exposure to cost overruns or delays. Additionally, they support the integration of digital twins and 360° walkthroughs, enabling the comparison of the as-built state with the planned model and the identification of millimetric discrepancies, as well as automatic alerts that link quality, cost, and schedule performance [133–135].

In the Strategic Decision and Communication domain, the integration of specialized AI tools reinforces key management competencies. Decision-Making (0.64) is strengthened by probabilistic prediction engines such as nPlan, whose algorithms analyze millions of historical activities to forecast delays and recommend the course of action with the lowest risk [136–139]. Strategic Thinking (0.60) benefits from the same scenario-based analyses, which project long-term impacts on portfolios and supply chains. Effective Delegation (0.57) becomes more precise through the integration of objective productivity metrics captured by Doxel, which compares the daily as-built progress with the plan and indicates which crews or equipment should be reassigned to avoid bottlenecks [140–142]. Effective Communication (0.55) is enhanced by OpenSpace.ai, whose 360° documentation linked to the BIM model provides geolocated images that management assistants transform into visual reports easily understood by clients, subcontractors, and field personnel [143,144]. Finally, Critical Thinking (0.53) is sharpened by platforms such as Newmetrix and Reconstruct, which respectively detect safety indicators through computer vision and compare actual progress with 4D simulations, providing evidence that challenges assumptions and promotes continuous reevaluation of success criteria [145,146].

Technical and Digital Skills is strengthened by artificial intelligence as it redefines professional practice and poses critical challenges such as BIM Utilization (0.78), which moves beyond mere software handling when automatic model verification is introduced through platforms like Avvir, which contrast point clouds with the BIM and generate issues in very short periods of time; however, the accuracy of the algorithm depends on high-density scans and a well-structured federated model, which exposes shortcomings in data standardization [147,148]. Digital Tools Proficiency (0.71) expands with the use of contextual assistants, as presented by the predictive functions of Generative Design and Autodesk Assistant in Revit, which automate the search for information, the generation of variants, and the initial construction of schedules, thereby freeing up time for higher-value tasks. In addition, they suggest families, critical paths, and construction sequences, accelerating workflows, although at the cost of a learning curve that may be rejected by some professionals who are less familiar with cloud environments [149–151].

The capacity for Technological Innovation (0.67) is driven by generative engines that propose parceling schemes or optimized structural solutions; however, the engines require well-justified input criteria, and their adoption may be hindered by organizational resistance to perceived risk. Material Knowledge (0.60), intelligent libraries such as Cove. Tool or One Click LCA estimate environmental impact and life cycle cost, enriching technical selection; however, the validity of their recommendations depends on local price databases and standards that still present gaps in emerging markets, in terms of data quality, platform interoperability, and critical capacity [152–154]. Finally, Blueprint Interpretation (0.49) is streamlined with vision engines, which extract dimensions, symbols, and discrepancies between disciplines; however, the reliability of these systems degrades with low-resolution scanned drawings, requiring human verification. Thus, AI amplifies each technical skill, but its final effectiveness is conditioned by the professional's ability to validate results and manage the cultural change implied by its adoption [155–158].

In the Compliance and Environmental Management domain, AI serves as an accelerator; however, its effectiveness depends on the quality of the data and human judgment. Regulatory compliance (0.76) is strengthened through assistants like UpCodes Copilot and field platforms such as HammerTech, which link codes and permits in real-time, although their scope is limited to the jurisdictions covered [159,160]. Environmental management (0.66) becomes more proactive through predictive systems, such as Envirosuite, and eco-optimization engines, like cove.tool, which are capable of anticipating dust, noise, or embodied carbon, provided that reliable local sensors and databases are available [161–163]. Finally, document management (0.51), facilitated by an AI engine, classifies permits and contracts and alerts users to outdated versions; however, its accuracy declines if the team fails to maintain consistent metadata.

4.4. Comparison of perceptions between analysis groups

Each of the 35 competences was analyzed using a Mann-Whitney U test across 135 valid questionnaires to determine whether managers and students perceive the influence of AI on competencies in construction management differently. The analysis focused on the median ratings through three perspectives: (1) the use of AI tools in project management, (2) the experience level of the respondents, and (3) their current professional role within the construction sector. A two-tailed significance criterion of 5 % ($p \leq 0.05$) was applied. P-values above this threshold led to the retention of the null hypothesis, indicating that the medians between groups are statistically equivalent. Significant differences were observed for Leadership (C_1), Risk Management (C_8), Effective Communication (C_{21}), and others (see Appendix B). This suggests that exposure to AI alters perceptions of these competencies in a group-specific manner. Items highlighted in red indicate $p \leq 0.05$, while those shaded orange (with p-values between 0.05 and 0.20) represent borderline effects that, although not conventionally significant, should be considered in future research. Competencies displayed in white showed no meaningful differences between groups, indicating a consensus on their importance for AI-enabled project success.

4.4.1. Perceptions analysis by use frequency of AI tools in construction management

Based on the reported frequency of AI tool usage in construction project management, a Mann-Whitney U test was conducted to compare two study groups. Group 1 consisted of managers and students with low levels of AI use, while Group 2 comprised participants with medium to high levels of AI engagement (see Table 2). The analysis revealed statistically significant differences in several core competencies. For Conflict Management (C_4) ($p = 0.00527$), Group 1 had a mean score of 3.14, whereas Group 2 averaged 3.55. This suggests that higher exposure to AI is correlated with an enhanced ability to address and resolve conflicts, likely because digital tools help facilitate dispute tracking, stakeholder communication, and resolution modeling. Regarding Planning (C_6) ($p = 0.01834$), Group 1 reported a mean score of 3.97, compared to 4.30 for Group 2. This difference indicates that more frequent users of AI perceive planning as more accurate and responsive, thanks to AI's capacity to process large datasets, simulate scenarios, and optimize resource allocation. In the area of Cost Control (C_7) ($p = 0.00056$), Group 1 had an average score of 3.79, while Group 2 achieved 4.25, highlighting a substantial gap. This reflects the advantage that AI-savvy professionals have in tracking expenditures in real time, identifying early deviations, and implementing corrective financial measures with greater efficiency.

Further analysis of group-level differences revealed statistically significant variations in several key competencies based on participants' reported levels of AI usage in construction project management. In Risk Management (C_8) ($p = 0.00786$), Group 1 (low AI use) had an average score of 3.83, while Group 2 (medium/high AI use) scored 4.18. This indicates that participants with greater exposure to AI perceive it as a

valuable tool for early risk identification and for implementing preventive measures through predictive modeling and scenario analysis. Similarly, in Change Management (C_9) ($p = 0.00216$), Group 1 averaged 3.67 compared to 4.05 for Group 2, suggesting that those more familiar with AI recognize its utility in rapidly adapting project plans and workflows in response to changing conditions. In Quality Control (C_{10}) ($p = 0.04005$), Group 1 scored 3.59 and Group 2 scored 3.79, reflecting the impact of AI-driven inspection tools and sensor-based monitoring on improving quality assurance processes. The competency of Blueprint Interpretation (C_{11}) ($p = 0.01062$) also showed a notable difference, with Group 1 averaging 3.47 compared to Group 2's 3.77. This highlights how AI-enabled visualization tools and interactive 3D model platforms enhance comprehension of complex design documentation. For Digital Tools Proficiency (C_{14}) ($p = 0.03254$), Group 1 averaged 3.81 while Group 2 scored 3.97, reinforcing the link between AI familiarity and the effective use of digital construction platforms. In Site Audits (C_{20}) ($p = 0.02662$), the gap between Group 1 (3.24) and Group 2 (3.56) suggests that AI adoption in on-site inspections (such as automated image analysis, drone-assisted data capture, and anomaly detection) contributes to more precise and efficient field verification. Finally, Effective Communication (C_{21}) ($p = 0.00004$) showed one of the strongest contrasts, with Group 1 averaging 3.50 and Group 2 averaging 4.05. This highlights the role of AI in enhancing communication through automated reporting, real-time dashboards, and integrated collaborative platforms that break down information silos, thereby expediting decision-making.

The comparative analysis revealed statistically significant differences in a set of high-order competencies related to decision-making, stakeholder interaction, and strategic leadership. In the area of Negotiation (C_{22}) ($p = 0.00527$), Group 1 reported a mean score of 3.26, while Group 2 achieved a score of 3.62. This suggests that AI tools enhance negotiation skills by providing real-time data, cost scenarios, and risk assessments that support more informed and assertive positions. For Customer Service (C_{24}) ($p = 0.03077$), Group 1 averaged 3.66, compared to 3.91 for Group 2. This indicates that AI-powered interfaces improve client engagement by automating request tracking, predicting inquiries, and streamlining communication across complex project environments. In Reporting (C_{25}) ($p = 0.02383$), Group 1 scored 4.02, while Group 2 scored 4.30. This shows that AI significantly enhances reporting quality by automating data aggregation and reducing human error, thereby facilitating more transparent and real-time project updates. Regarding Decision-Making (C_{26}) ($p = 0.02506$), Group 1 averaged 3.60, while Group 2 reached 3.95. This reflects how advanced analytics and AI-generated insights empower managers to make quicker, data-driven decisions. A similar trend was observed in Strategic Thinking (C_{28}) ($p = 0.03312$), where Group 1 scored 3.79 and Group 2 scored 4.05. This underscores that individuals with more AI experience are better positioned to anticipate disruptions and adjust long-term project goals using scenario planning and forecasting tools. In the Supply Chain Management category (C_{29}) ($p = 0.04666$), Group 1 had an average score of 3.62, while Group 2 scored 3.79. This difference highlights the significant role of AI in predicting material demand, optimizing logistics, and mitigating procurement delays. In the Professional Ethics category (C_{33}) ($p = 0.01217$), Group 1 achieved a score of 3.16 compared to Group 2's 3.48. This suggests that increased exposure to AI is associated with a greater awareness of ethical considerations such as algorithmic bias, data privacy, and responsible automation. For Resilience (C_{34}) ($p = 0.00363$), Group 1 reported a score of 3.17, while Group 2 scored 3.64. This indicates that individuals who utilize AI may be better equipped to adapt to unexpected disruptions and to reallocate resources dynamically. In the Results Orientation (C_{35}) ($p = 0.00437$), there was a notable divergence: Group 1 averaged 3.59 while Group 2 reached 3.97. This highlights AI's capacity to facilitate continuous performance monitoring and real-time progress tracking, thereby fostering a results-driven culture in project execution.

Although several competencies (such as Effective Delegation (C_3),

BIM Utilization (C_{12}), Material Knowledge (C_{13}), Technological Innovation (C_{15}), Occupational Safety (C_{16}), Environmental Management (C_{18}), Document Control (C_{19}), Supplier Management (C_{23}), Critical Thinking (C_{27}), Resource Optimization (C_{30}), Continuous Learning (C_{31}), and Adaptability (C_{32})) did not reach the predefined statistical significance threshold ($\alpha = 0.05$), many of their p-values fell within the range of $0.05 < p \leq 0.20$. This pattern suggests that these competencies may exhibit latent or emerging differences between individuals with low and high levels of AI tool adoption in construction project management. Such borderline trends warrant attention in future research, particularly in studies that aim to assess how increased interaction with AI technologies may gradually influence the perceived relevance or development of these managerial capabilities.

4.4.2. Perceptions analysis by experience in construction management

The Mann-Whitney U test was used to compare the level of industry experience among participants in two groups: Group 1, comprising participants with no prior experience in construction management, and Group 2, consisting of managers and students with relevant experience in construction management. The analysis revealed statistically significant differences in several key competencies, highlighting how professional background influences perceptions regarding the utility of AI in project management. For Leadership (C_1) ($p = 0.04080$), Group 1 reported a mean score of 3.47, while Group 2 averaged 3.10. This inverse pattern suggests that participants with industry experience may be more skeptical about AI's potential to enhance leadership, likely relying on interpersonal judgment and traditional management practices developed through on-site experience. In contrast, for Risk Management (C_8) ($p = 0.02859$), Group 1 had a mean score of 3.79, compared to 4.13 in Group 2. This suggests that experienced professionals place a higher value on AI's predictive capabilities for identifying and mitigating risks. Similarly, in Technological Innovation (C_{15}) ($p = 0.03997$), Group 1 scored 4.07, while Group 2 achieved a mean score of 4.35, indicating that those with field experience recognize AI's potential to streamline processes and facilitate the integration of emerging technologies. Regarding Strategic Thinking (C_{28}) ($p = 0.04809$), Group 1 averaged 3.72, while Group 2 averaged 4.03. This reflects a stronger appreciation among experienced participants for AI's ability to support long-term planning, scenario analysis, and proactive project alignment. These findings suggest that professional experience in construction enhances the recognition of AI's strategic value in specific areas, particularly in risk mitigation, innovation enablement, and forward-looking decision-making.

Several competencies, while not reaching the established threshold for statistical significance ($\alpha \leq 0.05$), demonstrated moderate differences between groups, with p-values falling within the range of $0.05 < p \leq 0.20$, suggesting potential areas for future research. These include Conflict Management (C_4), Blueprint Interpretation (C_{11}), BIM Utilization (C_{12}), Digital Tools Proficiency (C_{14}), Occupational Safety (C_{16}), Critical Thinking (C_{27}), and Adaptability (C_{32}).

In addition to these marginal trends, experience level notably affected how certain competencies were perceived. For example, experienced professionals placed greater value on AI's role in risk management and strategic thinking, while students tended to rate AI higher for leadership and teamwork competencies. Appendix B presents p-values for each competency by experience and role, highlighting where statistically significant differences emerged. These findings suggest that targeted training and implementation strategies may be needed to address the distinct needs of early-career professionals versus seasoned managers.

4.4.3. Perceptions analysis among students and managers

The final segment of the perception analysis involved a Mann-Whitney U test comparing two distinct groups of respondents: Group 1, comprising students enrolled in construction management courses, and Group 2, consisting of practicing construction managers. This

comparative assessment revealed statistically significant differences in some core competencies, highlighting how perceptions regarding key project management skills vary based on professional role and industry experience. For Leadership (C_1) ($p = 0.01141$), Group 1 reported a mean score of 3.39, while Group 2 averaged 2.55. This suggests that students perceive artificial intelligence as a significant enabler of leadership in construction projects, likely due to their exposure to digital collaboration tools, real-time analytics, and algorithm-assisted decision-making during their academic training. In contrast, experienced managers may rely more on interpersonal skills and traditional leadership practices shaped by their field experiences. Similarly, for Teamwork (C_2) ($p = 0.01125$), Group 1 averaged 3.49 compared to 2.62 for Group 2. This indicates that students place greater value on AI-supported collaboration tools (such as shared dashboards, automated coordination systems, and communication platforms). At the same time, managers may emphasize hands-on experience, hierarchical decision-making, and context-specific team dynamics. These results highlight a generational and experiential gap in perceptions of AI as a facilitator of managerial competencies, particularly in areas traditionally rooted in human interaction and organizational culture.

Risk Management (C_8) ($p = 0.00801$), Group 1 (students) reported a higher mean score of 3.85 compared to Group 2 (managers), which scored 3.45. This suggests that students view artificial intelligence as a powerful tool for identifying, predicting, and mitigating project risks. Their perspective is likely influenced by their academic exposure to AI-based tools such as machine learning models, scenario analysis, and predictive simulations. In contrast, managers, drawing from their field experience, often rely more on traditional risk management strategies and may be less inclined to recognize the high utility of AI in this context. When it comes to BIM Utilization (C_{12}) ($p = 0.01096$), students again rated their competence more favorably, scoring 3.85 compared to the 3.57 from managers. This difference reflects students' greater familiarity with digital modeling platforms, such as BIM, which are increasingly being incorporated into academic curricula. On the other hand, managers may encounter institutional barriers to adoption, including interoperability issues, cost constraints, and limited organizational readiness. In the realm of Technological Innovation (C_{15} , $p = 0.00008$), the disparity was even more pronounced, with students scoring 4.10 and managers scoring 3.79. Students view AI and digital tools as key drivers of transformation and increased productivity in the construction industry. In contrast, managers tend to take a more cautious approach, possibly due to concerns about implementation risks, training deficiencies, or resistance to change. Finally, for Adaptability (C_{32}) ($p = 0.00649$), students averaged a score of 3.68, whereas managers reported a score of 3.34. This suggests that students perceive adaptability as a crucial skill in a rapidly evolving construction environment characterized by digital disruption. Conversely, experienced managers may view adaptability as associated with incremental adjustments rather than systemic changes, often focusing on it from an operational standpoint.

Although the p-values did not meet the conventional threshold for statistical significance ($\alpha \leq 0.05$), a subset of competencies showed p-values between 0.05 and 0.20. This suggests moderate perceptual differences that deserve further investigation. The competencies in this range include Conflict Management (C_4), Mentoring (C_5), Planning (C_6), Cost Control (C_7), Quality Control (C_{10}), Material Knowledge (C_{13}), Digital Tools Proficiency (C_{14}), Occupational Safety (C_{16}), Environmental Management (C_{18}), Resilience (C_{34}), and Results Orientation (C_{35}). While these factors did not achieve statistical significance in the current Mann-Whitney U test, the variations observed in group means indicate nuanced perceptual gaps between construction managers and students enrolled in construction-related management courses that incorporate artificial intelligence tools.

These perceptual differences also suggest important implications for higher education curricula in construction management. Academic programs should evolve to incorporate sustained training in AI literacy,

data management, and evidence-based decision-making, complemented by explicit instruction in prompt design, model validation, and the critical interpretation of AI-generated results. Practical coursework should include integrated BIM and AI workflows for semantic search, scheduling support, and document intelligence through applied studios, capstone projects, and collaborative simulations that reflect real-world constraints. Interdisciplinary learning between engineering, data science, and management should be institutionalized through shared modules and co-assessed projects. Furthermore, ethical awareness, transparency, and accountability in AI use must become transversal learning outcomes, ensuring that future professionals are prepared to lead AI-enabled construction processes responsibly and effectively.

5. Discussion

AI adoption in construction is driven by the interaction between technological readiness and social, regulatory, and cultural factors that influence trust, risk perception, and attitudes toward innovation. Clear regulations on liability, data protection, and occupational safety determine investment levels and deployment speed, while procurement processes (such as award criteria, interoperability requirements, traceability, and auditability) either pose barriers or create opportunities for scale. Human capital, encompassing vocational training, university programs, and on-site upskilling, influences how technology is adopted and maintained. Supply-chain digital maturity, data standards, and information governance impact model quality and algorithm reliability. Organizational culture (including evidence-based practices, error tolerance, and learning mechanisms) and negotiations with insurers and unions shape the feasibility of new roles and protocols. Financial factors (cost of capital, return timelines, fiscal incentives), cybersecurity (vulnerability management, operational continuity), and public acceptance (perceived social value of automation) further influence results. When these factors align, AI advances from pilot projects to widespread use; when they diverge, benefits become fragmented, and sector-wide adoption slows.

In the United States, the ecosystem combines venture capital, academia-industry consortia, and standardization frameworks that facilitate controlled pilots and scaling; alternative delivery methods (e.g., design-build) and performance-based procurement provide opportunities for AI-driven solutions. However, market fragmentation, litigation risks, and state-level regulatory differences increase compliance costs and foster caution in public agencies, while data protection and cybersecurity raise governance challenges. In Chile, strong academic capacity and experience in technology-intensive sectors, such as mining and energy, support pilots and the development of BIM/analytics. Nevertheless, an SME-dominated industry, limited R&D spending, reliance on imported technology, and rigid public procurement slow the transition from pilots to full-scale operations. Currency volatility, regional connectivity gaps, and a shortage of hybrid talent combining site operations with data skills further contribute to bottlenecks. In Colombia, public digital initiatives coexist with low BIM maturity among contractors, informal supply chains, higher capital costs, and management cultures that prioritize lowest-price awards over performance. Leadership turnover in projects, dispersed data, and inconsistent enforcement weaken traceability and reduce data quality, which are essential for AI systems. Overall, these asymmetries reveal fundamental differences in financing, human capital, data governance, and contracting regimes, explaining why adoption progresses at varying speeds across these contexts.

The stratified analysis based on experience and role type delivers a comprehensive understanding of AI tool adoption and its perceived value within construction management. Professionals with extensive experience identified AI's capabilities in risk mitigation and fostering innovation, while students highlighted its benefits for facilitating collaboration and developing leadership skills. These variations highlight the necessity for customized AI training and integration approaches that address the distinct backgrounds and requirements of each

stakeholder group.

5.1. Challenges in AI adoption for construction management

While integrating AI into construction management marks a significant step toward digitalizing processes and optimizing decisions, its adoption faces structural, technical, and ethical challenges that must be critically addressed. The excitement surrounding AI's predictive and analytical abilities often masks the vulnerabilities involved in deploying these technologies within complex and fragmented environments, such as construction. These challenges impact not only model accuracy and data security but also institutional trust and workforce acceptance. Consequently, it is crucial to identify and manage risks related to data quality, algorithmic bias, cybersecurity, workforce resistance, and ethical issues. Recognizing these factors does not diminish AI's potential; instead, it emphasizes that its true value is realized only when technological innovation is supported by governance, transparency, and accountability. This section systematically explores the main challenges identified in both literature and practice, offering a balanced view of the opportunities and limitations involved in the digital transformation of the construction industry.

Data quality is crucial for AI tools to deliver value in construction management; however, the available information often varies, is incomplete, or mismatched across various sources, including BIM models, schedules, costs, site logs, and sensor data. In practice, capturing errors, incompatible formats, missing metadata, and inconsistent coding conventions weaken traceability and hinder analytical integration. These issues are compounded by fragmented workflows, where asynchronous updates and weak data governance lead to increased informational entropy and a higher risk of incorrect decisions. We propose three main strategies: (i) data governance frameworks with clear roles, validation policies, and metadata catalogs; (ii) standardized interoperability standards (e.g., IFC, ISO 19650) along with reproducible ETL/ELT pipelines and automated quality checks; and (iii) human-in-the-loop feedback from the field to identify outliers and improve labels. Only when the entire data lifecycle acquisition, curation, versioning, lineage, and quality control is formalized can AI depend on trustworthy "production datasets," reducing prediction variance and enabling reliable, auditable insights at project and portfolio levels. Algorithmic bias occurs when training data does not adequately represent different contexts, actors, or operational conditions, resulting in models that systematically favor or penalize certain practices, suppliers, or crews without a technical basis. In construction, this can appear as "optimal" sequencing that ignores local constraints or risk predictions that reflect past underinvestment in safety rather than actual causal factors. We address this with ex-ante fairness audits (demographic parity, equal opportunity), sensitivity diagnostics across subpopulations (greenfield vs. retrofit, buildings vs. infrastructure), and drift control to recalibrate models as conditions change. We also recommend model cards and data sheets to document assumptions, coverage, and limitations, along with mitigation techniques like re-weighting, targeted data collection, and adversarial learning. Cross-review among planning, safety, and cost teams supported by explainability tools (SHAP/LIME) helps uncover spurious variables and prevents biased outcomes. Framing bias as a techno-organizational risk reduces externalities, improves model generalization, and aligns AI with due diligence and compliance standards.

The proliferation of digital workflows (CDEs, cloud collaboration, digital twins, IoT) increases the attack surface of projects that contain sensitive technical, financial, and contractual data in BIM models, schedules, and estimates. Threats such as ransomware, IP exfiltration, version tampering, or unauthorized access can threaten operational continuity and project governance. We recommend a defense-in-depth approach that includes network segmentation and least-privilege access, encryption in transit and at rest, multi-factor authentication and strong identity management, signed version control and hashing for

integrity, and continuous monitoring (SIEM) with incident response plans. Contracts should incorporate cyber-resilience clauses (RTO/RPO, liabilities, and audits) and regular testing (including penetration tests and tabletop exercises). In multi-party environments, secure API-based interoperability and separation between training and production environments reduce the risk of propagation. Lastly, ongoing training for teams and vendors is essential, as most incidents stem from human factors (such as phishing and weak credentials); therefore, cybersecurity should be a core project control rather than just an IT issue. AI adoption often faces resistance due to concerns about job loss, loss of control, or increased administrative load, especially when tools are introduced without a clear purpose. To turn friction into engagement, we suggest a socio-technical approach: co-design use cases with supervisors, planners, and HSE staff; run targeted pilots with clear value metrics (rework, cycle time, incidents); and deliver quick wins that provide immediate benefits (fewer clashes, smoother coordination). Training should be localized and gradual, using micro-credentials and onsite "champions" as peer mentors. Change management requires transparent governance (executive sponsorship, process owners) and ongoing feedback to adapt interfaces and workflows to site needs. When AI is viewed as an ally that automates tedious tasks, predicts conflicts, and supports better decisions, technology anxiety diminishes and involvement increases, enabling sustainable gains in productivity and safety without sacrificing crew skills.

Finally, ensuring responsible AI in construction requires implementing ethical safeguards that address transparency, accountability, consent, and proportionality in the use of data and models. Use cases range from video analytics to crew risk scoring; without proper controls, organizations may face risks related to intrusive surveillance, opaque decision-making processes, or unintended harm. It is recommended to adopt responsible-AI frameworks that include operational principles, review boards, and pre-deployment ethical impact assessments. Explainability should be a core requirement: decision logs, data lineage, and access to justifications are vital for audits and appeals. Data minimization and purpose limitation are crucial, along with clear restrictions on reuse. Contracts need to define shared responsibilities among owners, contractors, and technology providers. Additionally, incorporating worker perspectives and occupational safety considerations into use-case design enhances social legitimacy and reduces reputational risks, aligning innovation with principles of rights, dignity, and safety. The adoption of AI in construction requires a comprehensive approach that extends beyond technical performance to include governance, evidentiary support, and social legitimacy. The value of models depends on trustworthy, traceable, and well-managed data; mechanisms to identify and correct biases; and secure infrastructures capable of withstanding incidents while maintaining project integrity. Workforce acceptance is strengthened through co-designed use cases, tangible on-site benefits, and targeted training that protect practical expertise. The ethical aspect also involves explainability, clear boundaries on data use, and shared responsibilities among owners, contractors, and technology vendors. Embedding these principles into verifiable operational frameworks transforms AI into a reliable tool that delivers consistent, auditable results aligned with productivity and safety goals. Digital maturity is judged less by algorithm complexity than by an organization's ability to implement practices with discipline, transparency, and continuous learning. When these principles are embedded into processes and contracts, AI facilitates faster decision-making, reduces uncertainty, and improves portfolio resilience.

5.2. Differences between academic and industry perspectives

While there is broad consensus on the potential of AI to enhance construction management competencies, our findings reveal important differences between academic and industry perspectives. Students, who are exposed to AI tools within updated curricula, tend to view these technologies as transformative for leadership, teamwork, and

adaptability. In contrast, practicing managers, drawing on field experience, are more cautious and emphasize the value of AI for risk management, strategic planning, and technological innovation. These divergences likely reflect differences in exposure, organizational culture, and practical constraints. For example, students may be more familiar with collaborative digital platforms and theoretical applications, while managers must contend with implementation barriers such as data quality, interoperability, and workforce resistance. The generational gap in perceptions suggests that educational programs should not only focus on technical skills but also address real-world challenges faced by industry professionals. To bridge these gaps, academic curricula should incorporate practical case studies, industry partnerships, and hands-on experience with AI tools. Conversely, industry training programs should emphasize continuous learning, change management, and the integration of AI into existing workflows. By aligning educational and professional development strategies, both academia and industry can better prepare future managers to leverage AI effectively in diverse construction environments.

5.3. Limitations and future work

This study collected perceptions from two specific groups: project managers and students in related fields. These participants were selected through purposive sampling, which aimed to highlight the differing viewpoints between practicing professionals and those still in training. As a result, other essential stakeholder groups in the project life cycle, such as contractors, subcontractors, site and field engineers, supervisors, owner's representatives, regulators, oversight bodies, clients, and developers, were not included. This limitation affects the representativeness and external validity of the findings. This decision aligns with the exploratory goal of assessing how AI assistants might enhance management competencies from these two distinct educational backgrounds. However, it outlines an important agenda for future research that should encompass a broader range of stakeholders and project phases. Such research should employ stratified or quota sampling designs and integrate mixed methods, combining surveys with interviews, focus groups, and on-site observations to capture more contextual and behavioral nuances beyond self-reports. Additionally, future studies should aim to include observable performance outcomes and utilize longitudinal designs, along with weighting and post-stratification procedures, to enhance generalizability. More information about the two target populations and the purpose of the sampling strategy can be found in the Methods section.

Building on the exploratory nature of this study, future research could utilize longitudinal and case-based approaches to gain a deeper understanding of how AI adoption influences managerial competencies over time and in real construction environments. While this study focused on perceptions, it did not capture the behavioral or performance-based evidence of AI integration in practice. Longitudinal studies could track how managers and multidisciplinary teams evolve as AI tools become integral to their daily workflows, examining changes in decision-making quality, collaboration dynamics, and productivity indicators. Additionally, in-depth case studies on active construction projects could analyze the operational mechanisms through which AI assistants contribute to planning accuracy, resource optimization, safety management, and knowledge retention. Such approaches would enable researchers to move beyond perception analysis toward measuring actual improvements in performance and organizational transformations resulting from AI-enabled management. Moreover, comparative case studies across different project types, regions, or levels of technological maturity could illuminate the contextual factors that influence AI adoption and its effectiveness. Ultimately, these methodological advancements would enhance the explanatory power of future research by connecting perceptual insights with empirical evidence, thereby contributing to a more comprehensive understanding of how AI-driven assistants reshape managerial competencies, work practices, and

learning processes within the construction industry.

6. Conclusions

This study expands existing theoretical knowledge in four key areas. First, it identifies 35 core competencies required for effective construction project management through a systematic review of the literature. These competencies span across planning, execution, monitoring, and closure phases, highlighting the diverse skill set needed for successful project management. Second, it analyzed the perceptions of construction managers and students regarding the impact of AI on these competencies. The top five managerial competencies that are most influenced by AI tools are: (1) technological innovation (C_{15} ; $RII = 0.846$), (2) planning (C_6 ; $RII = 0.834$), (3) reporting (C_{25} ; $RII = 0.831$), (4) document control (C_{19} ; $RII = 0.803$), cost control (C_7 ; $RII = 0.803$). Third, it established six AI-influenced competency domains derived from Exploratory Factor Analysis (EFA): leadership and collaborative management, adaptability and results orientation, project control and risk management, strategic decision-making and communication, technical and digital skills, and compliance and environmental management. Identifying these AI-influenced competency domains provides a theoretical framework for understanding and leveraging AI to enhance project management competencies, addressing a critical gap in construction management education and practice. Unlike previous research efforts, this study adopts an interdisciplinary perspective that integrates technological innovation, operational efficiency, and strategic planning to determine the competencies needed. Four, the study examined differences in perceptions of AI's impact between construction professionals and students, considering the applicability and benefits of AI tools. Practical experience and theoretical knowledge influence these perceptions, underscoring the need for competency development strategies aligned with industry realities. Understanding how perceptions vary based on experience, professional role, and educational background provides information for tailoring training programs, ensuring that proposed measures are theoretically sound and operationally feasible.

The practical contributions derived from the theoretical aspects presented in this study highlight how AI can enhance the competencies of construction managers and students. Recognizing the most impactful areas, such as technological innovation, planning, reporting, document control, and cost control, enables the prioritization of training and development efforts, ensuring that resources are efficiently allocated to the most beneficial areas. For construction companies, these findings provide information for improving training programs and integrating AI tools into project management practices, ensuring that managers and students are equipped with the necessary skills to leverage AI effectively. Integrating these competency development strategies into professional training can enhance project outcomes, minimize disruptions, and improve overall efficiency.

From the perspective of construction managers and educators, this study offers objective criteria for assessing competency development needs, identifying gaps, and establishing more effective training mechanisms. Moreover, the categorization of competencies by impact allows for the development of targeted training protocols tailored to the actual needs of the construction sector rather than relying on generic training measures that may be less effective. At an industry-wide level, adopting these strategies enhances project management competencies, improves operational efficiency, strengthens corporate reputation, and fosters a more innovative work environment. Consequently, this study advocates for the development of training programs to promote a stronger culture of competency development, particularly in construction management, where professionals are exposed to various operational challenges. To achieve this, it is advisable to develop a comprehensive strategy for integrating AI-driven platforms into project scheduling, cost estimation, and risk assessment processes. This approach should be complemented by initiatives that advance workforce development in AI competencies

and data analysis.

Integration of these practices requires a holistic approach. In addition to industry adoption, a strategy is also needed to integrate AI into education and policies. Universities are encouraged to incorporate AI competencies into their curricula, adding modules that address ethical practices, data analysis, and prompt generation. Policymakers can also support this effort by developing standardized frameworks for AI skill recognition, establishing ethical compliance, and offering incentives for the responsible use of AI and its adoption.

This study presents four key limitations, which can be addressed in future research. First, the analysis was based solely on data collected from construction managers and students, potentially introducing a bias by excluding the perspectives of other key stakeholders, such as field workers and specialized trades. These individuals, who are directly involved in project execution, could provide valuable insights into the practical application of AI tools. Second, the cultural and socioeconomic conditions in different regions introduce contextual factors that may influence perceptions and adoption of AI. The assessment of AI's impact can vary between urban and rural areas, due to differences in infrastructure, resource availability, and regulatory enforcement. Third, although this study employs a methodology for evaluating AI's impact on competencies, it does not develop a comprehensive implementation guide.

Future research could explore how AI tools can be integrated into both the early planning phase and the execution stage of construction projects, ensuring that competency development measures are tailored to each phase. Fourth, this study focuses exclusively on construction management, excluding other types of infrastructure projects, such as roads and bridges, which present distinct challenges and require different competency development strategies. For future research directions, it is recommended to expand the scope of analysis by incorporating the perspectives of a broader range of industry stakeholders, ensuring a more comprehensive and representative assessment of AI's impact on project management competencies. Additionally, investigating how cultural and socioeconomic differences impact perceptions and adoption of AI across various regions would provide further insights. Finally, the implementation of digital methodologies, such as BIM, emerges as a key opportunity for enhancing competency development. BIM enables more precise planning and facilitates the design of practical solutions based on detailed simulations and in-depth analysis throughout all project phases.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used *Grammarly* in order to improve readability, spelling and grammar. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. AI-influenced competency domains for construction management

Domains	Id	Competencies to manage construction projects	Main Components					
			1	2	3	4	5	6
Leadership and Collaborative Management	C ₁	Leadership: Guide and motivate the team to achieve project objectives.	0.83					
	C ₂	Teamwork: Promote collaboration among architects, engineers, and contractors.	0.78					
	C ₃₃	Professional Ethics: Act with honesty and transparency in all decisions.	0.64					
	C ₂₂	Negotiation: Reach mutually beneficial agreements with suppliers and contractors.	0.59					
	C ₁₆	Occupational Safety: Ensure a safe working environment for all personnel.	0.58					
	C ₂₀	Site Audits: Conduct inspections to verify compliance with standards.	0.56					
	C ₄	Conflict Management: Resolve disputes between workers or contractors in a fair manner.	0.56					
	C ₂₃	Supplier Management: Coordinate the delivery of materials and services.	0.52					
	C ₂₉	Supply Chain Management: Ensure timely procurement of materials.	0.37					
Adaptability and Results Orientation	C ₃₂	Adaptability: Adjust to changes in project scope or external conditions.		0.67				
	C ₅	Mentoring: Train and develop skills in personnel.		0.67				
	C ₂₄	Customer Service: Respond to client needs and expectations.		0.65				
	C ₃₁	Continuous Learning: Stay up-to-date with new construction technologies and methods.		0.63				
	C ₃₄	Resilience: Remain calm and productive in crisis situations.		0.59				
	C ₃₀	Resource Optimization: Maximize efficiency in the use of labor and equipment.		0.47				
	C ₃₅	Results Orientation: Focus on efficiently achieving project objectives.		0.45				
Project Control and Risk Management	C ₈	Risk Management: Identify and mitigate risks that may affect the project.				0.76		
	C ₁₀	Quality Control: Ensure materials and processes meet required standards.				0.69		

(continued on next page)

(continued)

Strategic Decision and Communication	C ₉	Change Management: Adapt the project in response to unexpected modifications.	0.65	0.64
	C ₆	Planning: Develop detailed work plans with defined timelines and resources.	0.65	
	C ₇	Cost Control: Monitor and adjust the budget to avoid cost overruns.	0.54	
	C ₂₅	Reporting: Prepare clear reports on project progress.	0.47	
	C ₂₆	Decision-Making: Evaluate alternatives and choose the best option for the project.		
Technical and Digital Skills	C ₂₈	Strategic Thinking: Plan long-term strategies for project or business growth.	0.60	0.53
	C ₃	Effective Delegation: Assign appropriate tasks to team members based on their skills.	0.57	
	C ₂₁	Effective Communication: Convey clear information to clients, teams, and suppliers.	0.55	
	C ₂₇	Critical Thinking: Analyze data and situations to make informed decisions.		
	C ₁₂	BIM Utilization: Operate Building Information Modeling tools.	0.78	
	C ₁₄	Digital Tools Proficiency: Use specialized software (e.g., AutoCAD, MS Project, BIM).	0.71	
	C ₁₅	Technological Innovation: Implement new technologies in project development.	0.67	
Compliance and Environmental Management	C ₁₃	Material Knowledge: Select appropriate materials for each phase of the project.	0.60	0.76
	C ₁₁	Blueprint Interpretation: Read and understand blueprints from various project areas.	0.49	
	C ₁₇	Regulatory Compliance: Ensure the project complies with local laws and regulations.		
	C ₁₈	Environmental Management: Minimize the project's environmental impact (e.g., waste reduction).	0.66	
	C ₁₉	Document Control: Organize and manage permits, contracts, and warranties.	0.51	

Appendix B. Mann-Whitney *U* test *p*-values by analysis group

Id	Competencies to manage construction projects	Use	Experience	Role
C ₁	Leadership: Guide and motivate the team to achieve project objectives.	0.42833	0.04080	0.01141
C ₂	Teamwork: Promote collaboration among architects, engineers, and contractors.	0.60026	0.26490	0.01125
C ₃	Effective Delegation: Assign appropriate tasks to team members based on their skills.	0.14940	0.89248	0.65766
C ₄	Conflict Management: Resolve disputes between workers or contractors in a fair manner.	0.00527	0.17669	0.12338
C ₅	Mentoring: Train and develop skills in personnel.	0.58102	0.71231	0.05526
C ₆	Planning: Develop detailed work plans with defined timelines and resources.	0.01834	0.78771	0.13696
C ₇	Cost Control: Monitor and adjust the budget to avoid cost overruns.	0.00056	0.40705	0.17326
C ₈	Risk Management: Identify and mitigate risks that may affect the project.	0.00786	0.02859	0.00801
C ₉	Change Management: Adapt the project in response to unexpected modifications.	0.00216	0.76416	0.24617
C ₁₀	Quality Control: Ensure materials and processes meet required standards.	0.04005	0.44065	0.13440
C ₁₁	Blueprint Interpretation: Read and understand blueprints from various project areas.	0.01062	0.11566	0.24812
C ₁₂	BIM Utilization: Operate Building Information Modeling tools.	0.17366	0.10070	0.01096
C ₁₃	Material Knowledge: Select appropriate materials for each phase of the project.	0.10121	0.25672	0.14691
C ₁₄	Digital Tools Proficiency: Use specialized software (e.g., AutoCAD, MS Project, BIM).	0.03254	0.18159	0.13739
C ₁₅	Technological Innovation: Implement new technologies in project development.	0.18504	0.03997	0.00008
C ₁₆	Occupational Safety: Ensure a safe working environment for all personnel.	0.05709	0.05247	0.19152
C ₁₇	Regulatory Compliance: Ensure the project complies with local laws and regulations.	0.52938	0.34978	0.79980
C ₁₈	Environmental Management: Minimize the project's environmental impact (e.g., waste reduction).	0.14508	0.24436	0.15946
C ₁₉	Document Control: Organize and manage permits, contracts, and warranties.	0.06746	0.34723	0.29544
C ₂₀	Site Audits: Conduct inspections to verify compliance with standards.	0.02662	0.76866	0.79463
C ₂₁	Effective Communication: Convey clear information to clients, teams, and suppliers.	0.00004	0.98707	0.94215
C ₂₂	Negotiation: Reach mutually beneficial agreements with suppliers and contractors.	0.00527	0.58040	0.81834
C ₂₃	Supplier Management: Coordinate the delivery of materials and services.	0.10099	0.69471	0.29443
C ₂₄	Customer Service: Respond to client needs and expectations.	0.03077	0.65032	0.35062
C ₂₅	Reporting: Prepare clear reports on project progress.	0.02383	0.25644	0.00408
C ₂₆	Decision-Making: Evaluate alternatives and choose the best option for the project.	0.02506	0.26700	0.03701
C ₂₇	Critical Thinking: Analyze data and situations to make informed decisions.	0.15548	0.18779	0.00656
C ₂₈	Strategic Thinking: Plan long-term strategies for project or business growth.	0.03312	0.04809	0.01835
C ₂₉	Supply Chain Management: Ensure timely procurement of materials.	0.04666	0.24175	0.04836
C ₃₀	Resource Optimization: Maximize efficiency in the use of labor and equipment.	0.12426	0.69061	0.03346
C ₃₁	Continuous Learning: Stay up-to-date with new construction technologies and methods.	0.09810	0.45885	0.04919
C ₃₂	Adaptability: Adjust to changes in project scope or external conditions.	0.09998	0.12031	0.00649
C ₃₃	Professional Ethics: Act with honesty and transparency in all decisions.	0.01217	0.27514	0.79101
C ₃₄	Resilience: Remain calm and productive in crisis situations.	0.00363	0.48738	0.19015
C ₃₅	Results Orientation: Focus on efficiently achieving project objectives.	0.00437	0.75663	0.17561

Data availability

Data will be made available on request.

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